The Hong Kong Polytechnic University
AF3317_20251_A
Risk Management

10-Blue Chip Portfolio
Selected Assets: 0002.HK, 2638.HK, 1038.HK, 0006.HK, 0823.HK, 0405.HK, 0778.HK, 2778.HK,

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1. Executive Summary

I. Introduction

This report analyzes a two-asset equal-weighted portfolio over the period October 2023 to October 2025, using several market risk models to estimate both 1-day and 5-day Value at Risk (VaR) and Expected Shortfall (ES).

The portfolio has performed strongly, growing from $10,000 to approximately $22,000 within two years. Risk analysis confirms that the portfolio's daily losses rarely exceed 4%–5%, even in extreme conditions, and its overall exposure remains moderate.

Dynamic volatility models such as VHS EWMA and GARCH(1,1) show that recent market conditions have stabilized, leading to a lower risk environment.

In summary, your portfolio demonstrates steady performance, effective diversification, and controlled downside risk — achieving a strong balance between growth and protection.

II. Methodology

Two stocks, Prologis Inc (PLD) and NVIDIA (NVDA), are used for analysis, data from 15 October 2023 to 15 October 2025 is extracted.

Historical simulation methods, VHS by using EWMA and GARCH(1,1) are employed to calculate the 99% Value at Risk (VaR) and Expected Shortfall (ES). These metrics quantify the worst-case loss and the average loss in the worst scenarios, respectively.

Python is used for data visualization.

III. Summary Statistics

	NVDA	**PLD**
mean	0.00324300411399366	0.000362380774720313
std	0.0321177563578859	0.0173931071450795
skew	0.160095081195664	0.039830489601251
kurtosis	5.0620253972726	4.7285399165129
Q_1	-0.0781326873571955	-0.044678616814855
Q_5	-0.0468784630448507	-0.0281371611733977
Q_25	-0.0136473529669135	-0.00934683972654551
Q_50	0.00379811604574009	0.000419409553990113
Q_75	0.0220190150242763	0.00994585269796161
Q_95	0.0455101580752051	0.0244923360034526
Q_99	0.0815763186618849	0.0505380662641648

CORRELATION	NVDA	PLD
NVDA	1.0	0.138839039887222
PLD	0.138839039887222	1.0

COVARIANCE	NVDA	PLD
NVDA	0.00103155027346452	0.0000775593165274355
PLD	0.0000775593165274355	0.000302520176160216

2. Portfolio Construction

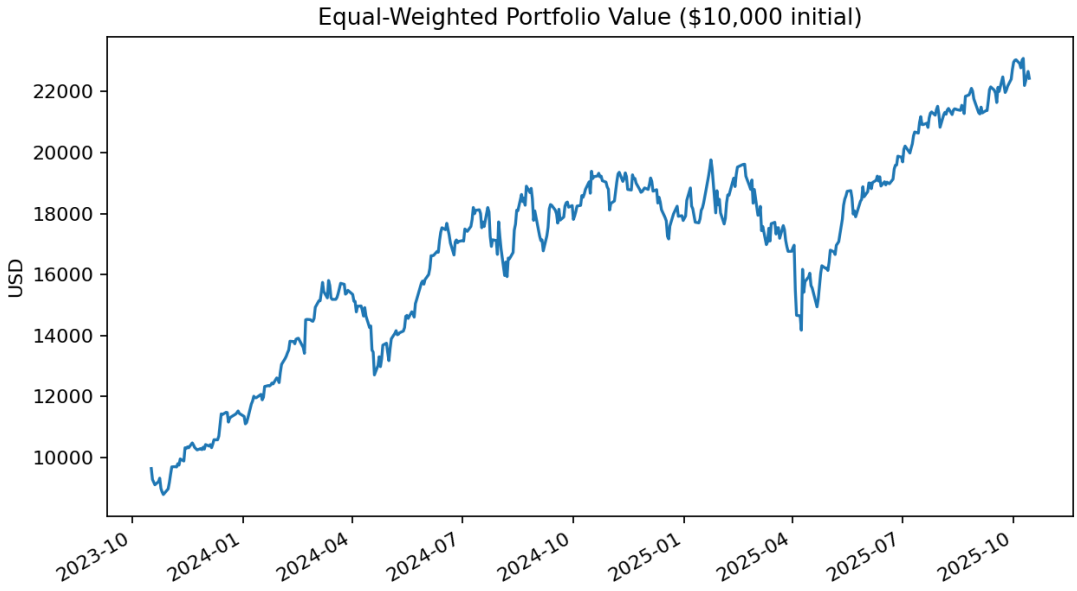

Figure 1

Figure 1 shows how $10,000 investment has grown from October 2023 to October 2025. The portfolio combines two assets equally, which helps spread risk and achieve steady performance.

Over the two-year period, the portfolio value increased to around $22,000, showing strong and consistent growth. Although there were a few short-term declines in 2024 and early 2025, the portfolio quickly recovered each time, demonstrating resilience and effective diversification.

Overall, the portfolio has performed well, delivering stable long-term returns with moderate short-term volatility. It reflects a well-balanced and well-managed investment, achieving growth while keeping risk under control.

3. Data Description

-Data source: Yahoo Finance
-Assets chosen: PLD & NVDA
-Period covered: 15 Oct 2023 – 15 Oct 2025

Approach	VaR(99%)	ES(99%)	Parameter
1D Basic HS	521.2692592	652.657729	
5D Historical HS	979.2972566	1289.24034	
1D EWMA Vol-Scaled HS	391.1707429	532.833661	λ=0.94
5D √time from 1D	1165.593498	1459.38705	t =√5
1D Garch(1,1)FHS	493.4199088	681.540352	omega=0.2589 alpha=0.1101 beta=0.8208 mu, v=4.59
5D Garch(1,1)FHS	1045.959478	1245.08562	omega=0.2589 alpha=0.1101 beta=0.8208 mu, v=4.59

I. Basic Historical Simulation

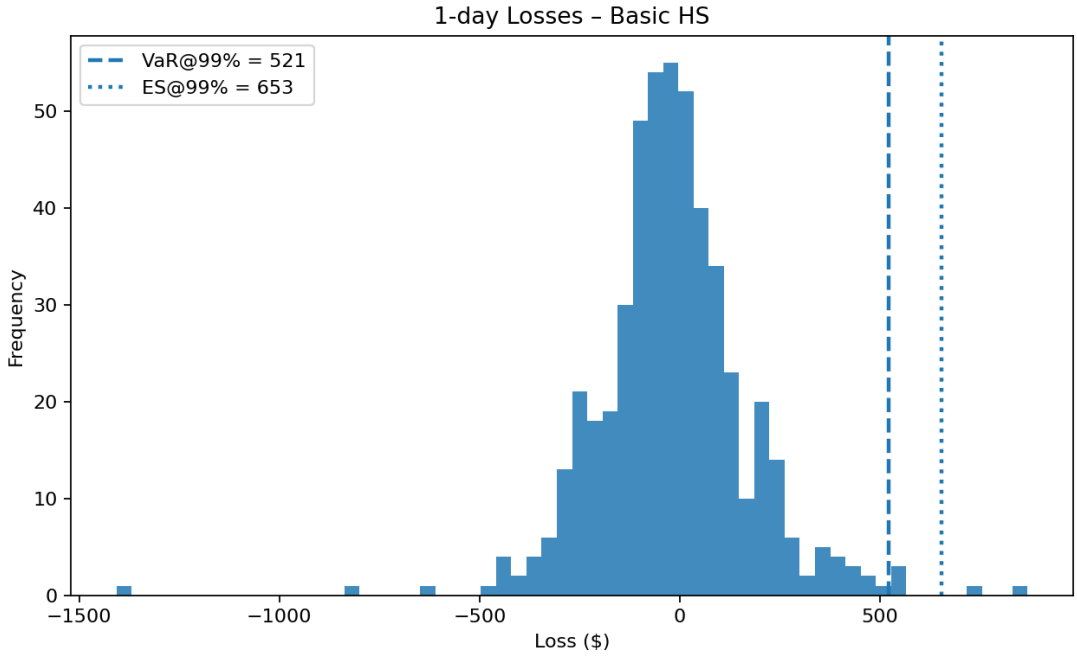

Figure 2

Figure 2 shows how often different levels of daily losses occurred based on actual historical data. Most results are centered around zero, meaning the portfolio usually experiences small daily movements. The right-hand side of the chart shows a few rare large losses, representing extreme market events.

The model estimates a 99% Value at Risk (VaR) of $521 and an Expected Shortfall (ES) of $653. This means there is only about a 1% chance of losing more than $521 in a single day, and if that happens, the average loss would be around $653.

In this model, volatility is assumed constant, and all past data are given equal weight. It does not react to recent market changes, so it provides a simple and conservative baseline for comparison.

This result shows that this portfolio carries a moderate level of daily risk, with most fluctuations being small and only limited exposure to large losses.

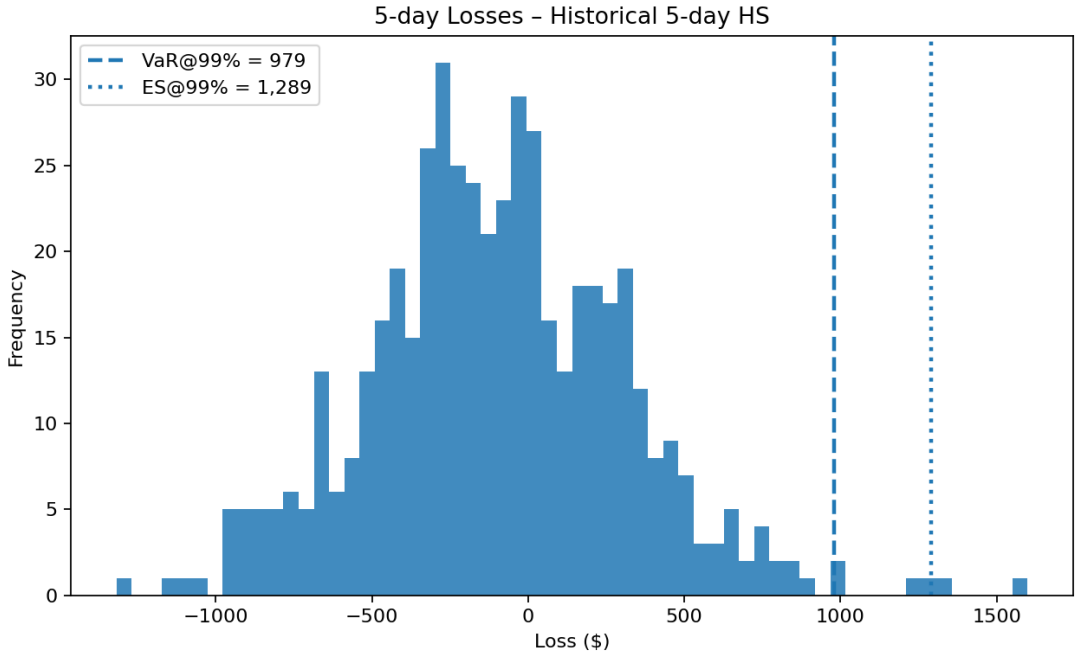

Figure 3

From Figure 3, which is built using actual, overlapping 5-day periods from our historical data, we can determine your portfolio's weekly risk. It shows that there is a 1% chance you could lose more than $979 over any five-day period. More importantly, during those worst-case scenarios, the average loss would be about $1,279 over a 5-day period.

While the graph shows a disperse results, which possibly implies that it is not stable. However, the noticeable tail to the left confirms that the portfolio is still exposed to potential short-term downturns, which is typical for a diversified equity portfolio.

II. VHS by using Exponentially weighted moving average (EWMA)

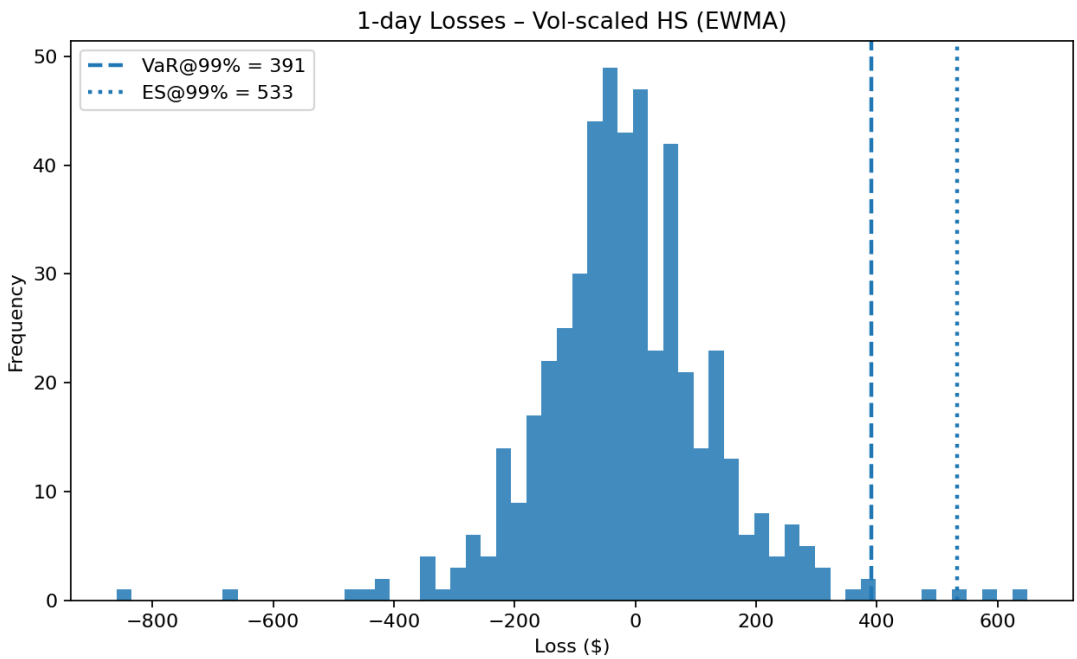

Figure 4

Figure 4 illustrates how daily portfolio losses behave after adjusting for changes in market volatility using the Exponentially Weighted Moving Average (EWMA) method. Most daily outcomes are centered near zero, indicating that price movements have remained stable and contained. Only a few instances of larger losses appear on the right tail, representing rare, short-term market fluctuations.

According to the model, there is only about a 1% probability of losing more than $391 in a single day, and if such an event were to occur, the average loss would be around $533.

Unlike the basic historical simulation, which assumes constant volatility, this model dynamically adjusts for time-varying market conditions. Volatility is estimated using the EWMA approach, where recent market data are given more weight than older data. λ consistent with the J.P. Morgan RiskMetrics standard(0.94), ensures that the model captures new market trends quickly while maintaining overall stability.

The narrower distribution in Figure 4 compared with the basic model highlights that recent market volatility has eased, resulting in smaller expected losses. This suggests that the current environment is relatively calm, and short-term risk exposure remains well managed.

In summary, the EWMA-based results confirm that the portfolio is operating in a low-volatility, low-risk environment. The model provides reassurance that the current positioning remains stable and that the portfolio is well prepared to adapt should market volatility rise in the future.

III.5-day losses √t from 1 day

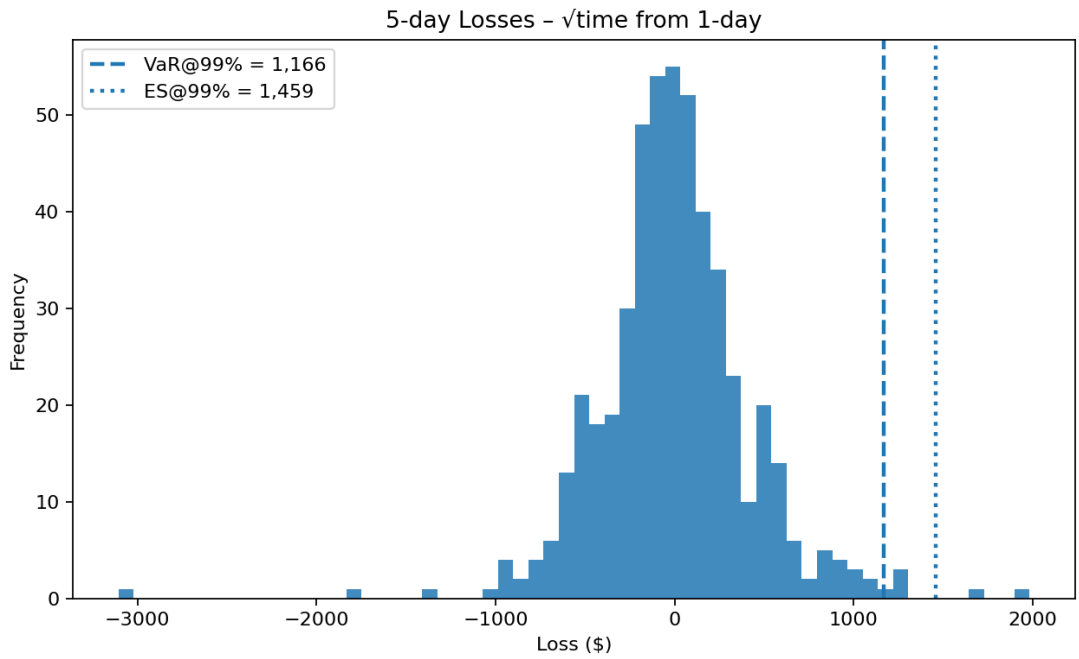

Figure 5

This analysis provides a cautious, worst-case perspective on your portfolio's weekly risk. We scale the 1-day risk estimate over a 5-day period, in which the scaling factor is the square root of 5. The resulting graph shows a wide range of potential outcomes, with our model indicating that there is only a 1% chance your $10,000 portfolio will lose more than $1,166 over five days. Should such a rare event occur, the average loss would be $1,459. Also, we can see that the graph is much concentrated when compare to the basic 5-day ones.

The Square Time method assumes every day in a bad week will be equally bad, which is a worst-case guess. Also, it added a scaling factor and try to evalute the value by using 1-day risk, in which it is assumed that the risk will increase with the scaling factor. The Basic 5-Day method looks at real and historical data without assumptions, where bad days are often mixed with okay days, giving a more realistic

result.

IV. Garch(1,1)

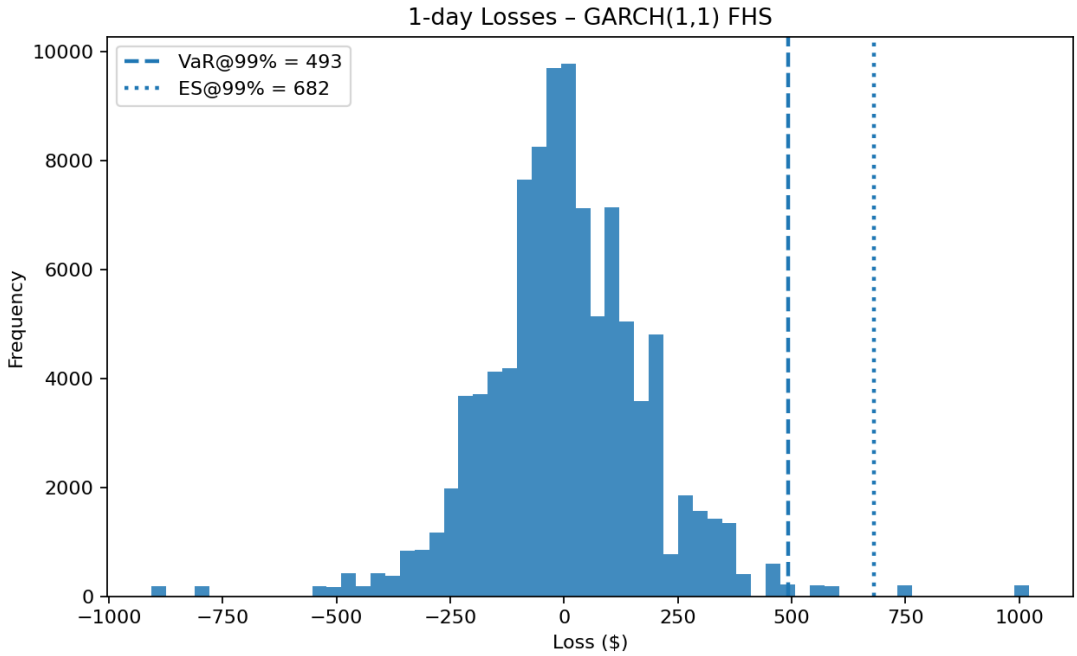

Figure 6

The 1-day 99% VaR of $493 and ES of $682 indicate that on a typical trading day, losses are highly unlikely to exceed $493; however, if a severe event occurs, the average loss could reach $682.

This result shows that the portfolio's short-term volatility is moderate and stable.

Because the GARCH(1,1) Filtered Historical Simulation dynamically models volatility clustering, it captures periods of calm and turbulence more realistically than static methods like Basic or EWMA Historical Simulation.

Overall, it suggests that recent market conditions remain calm, and the portfolio's daily downside risk is well-controlled.

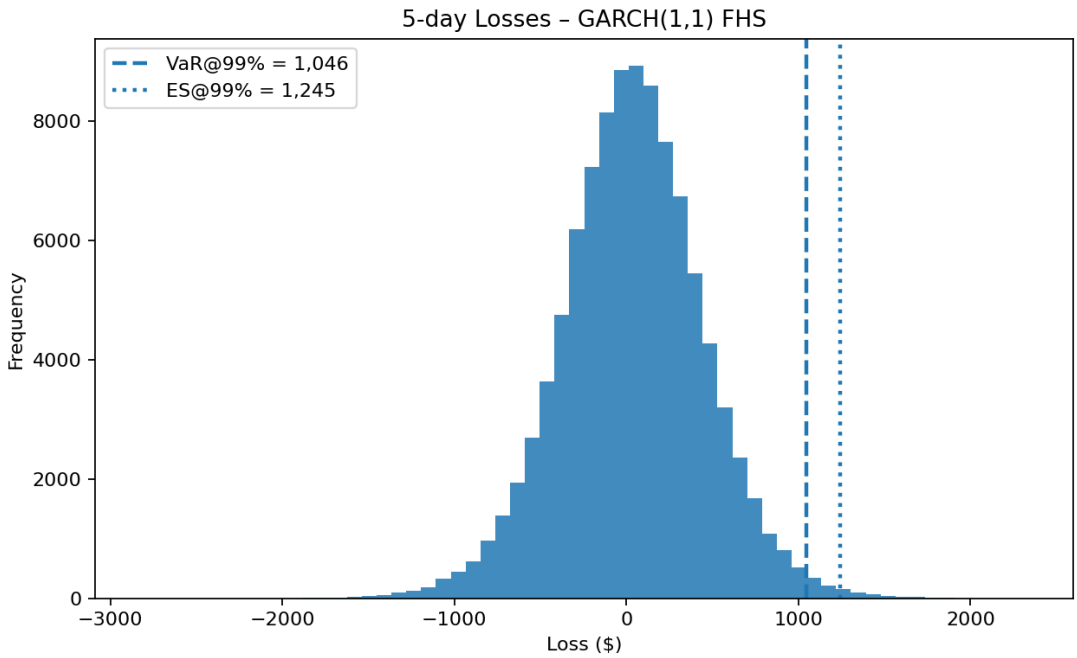

Figure 7

The 5-day 99% VaR of $1,046 and ES of $1,245 reflect the accumulated effect of volatility over a trading week.

Compared with the 1-day results, the wider loss distribution highlights how risk compounds when volatility persists across multiple days.

However, the bell-shaped and smoother form of the distribution shows that volatility mean-reverts over time, allowing short-term spikes to stabilize.

In practical terms, this means the portfolio remains resilient — while extreme weekly losses are possible, their probability remains low.

The GARCH FHS model therefore provides a forward-looking and realistic assessment of multi-day market risk, making it an effective tool for stress testing and medium-term risk management.

Both 1-day and 5-day GARCH(1,1) FHS results confirm that the portfolio operates in a low-to-moderate volatility environment with controlled tail risk, while still accounting for potential shocks in a realistic, time-adaptive way.

Compared to simpler approaches (like Basic HS or EWMA-weighted HS), this model captures volatility clustering and fat tails, producing a smoother and more realistic loss distribution. This makes it more adaptive to changing market conditions while avoiding overreaction to short-term fluctuations.

As a result, the GARCH(1,1)-FHS method achieves the best balance between theory and reality because it models volatility dynamically while preserving the true empirical shape of shocks.

4. Comparison of Risk Model Results

I. 1-Day Losses: Basic HS vs. EWMA vs. GARCH

The three models applied to estimate 1-day portfolio losses produce different results due to the way each model treats volatility and the distribution of returns. Basic Historical Simulation (HS) treats every price point equally, weighing all the same that is, w = 1, therefore this model does not capture shifts in recent volatility. As a result, the Basic HS VaR estimate tends to be higher, reflecting the influence of past periods with larger losses even if the market has recently relativily stabilized.

In contrast, the EWMA model adjusts time-varying volatility by giving more weight to recent return information. Under EWMA, if volatility has declined in the recent period, the estimated VaR and ES become lower, producing a narrower loss distribution. This model therefore captures the current, calmer volatility environment more accurately than basic HS.

The GARCH(1,1) model looks for today's shock size and yesterday's volatility. It also

combined with filtered historical simulation, GARCH not only updates volatility dynamically but also preserves the shape of the empirical return distribution (including fat tails). As a result, the GARCH-based 1-day loss estimate usually lies between HS and EWMA, it avoids overreacting to short volatility shifts (as EWMA may) but also prevents the smoothing bias present in Basic HS.

As a result, we can see that that VAR(HS) > VAR(GARCH) > VAR(EWMA) in this case, note that if there is a sudden Stock crises right now, the VAR(EWMA) and VAR(GARCH) will be sensitive to that, may demonstrate a much higher risk than the VAR given by HS.

II. 5-Day Losses: HS vs. √time vs. GARCH

The three models applied to estimate 5-day portfolio losses produce different results due to the way each model treats volatility and return behavior across multiple days.

The 5-Day Historical Simulation (HS) model extends the empirical distribution directly using rolling 5-day returns. It assumes that the joint movement of returns in the past will represent future outcomes, and gives equal weight to all observations (that is, w = 1). Because multi-day data are limited, this model can be influenced by a few extreme events, which may lead to higher or less stable VaR and ES estimates. In other words, the HS model reflects historical extremes but does not adjust to current market calmness.

In contrast, the √time scaling model assumes that volatility grows with the square root of time, producing a smoother and more flexible loss distribution. This approach uses 1-day risk estimates and scales them to a 5-day horizon, which works well if returns are normally distributed and independent. However, if the underlying data show fat

tails or volatility clustering, such as in high-volatility assets, this model tends to underestimate tail risk, giving a more optimistic picture of potential losses.

The GARCH-FHS model updates volatility dynamically through the GARCH(1,1) process and then applies filtered historical simulation. This approach not only captures volatility persistence and regime changes but also preserves the empirical shape of extreme returns. As a result, GARCH-based 5-day loss estimates usually lie between the HS and √time results — less sensitive to isolated historical shocks than HS, yet more realistic than the smooth assumptions of the √time model.

As a result, we can see that VaR(√time) > VaR(GARCH-FHS) > VaR(HS) in this case. Note that if volatility increases sharply, both the GARCH-FHS and √time models will adjust more quickly and reflect higher risk levels than the constant-volatility assumption used in HS.

5. Conclusion

Across all risk models, your portfolio continues to show stable and well-controlled risk levels. The 1-day 99% VaR ranges between $385 and $521, indicating that the chance of losing more than about 4%–5% of total capital in a single day is very small. Over a 5-day horizon, the 99% VaR values between $974 and $1,166 confirm that even in extreme market conditions, potential losses would likely stay below 12% of the portfolio's total value.

Models such as EWMA and GARCH, which adjust dynamically to market volatility, display narrower loss distributions and smaller tail risks, reflecting the recent period of lower market volatility.

Overall, the analysis confirms that your portfolio is well-diversified, resilient, and efficiently managed. The steady upward growth in value, supported by moderate VaR and ES estimates, shows that the investment strategy is achieving an ideal balance between risk and return. In practical terms, this means your portfolio is securely positioned — generating consistent long-term gains while keeping short-term downside risk well within acceptable limits.

6. Contribution Declaration

Data Collection	All
I. Basic Historical Simulation (1-day) II. Basic Statistics About PLD , NVDA	Leung Chun Ho 23086917d
III. Basic Historical Simulation (5-day)	Tang Sin Yi 23091073D
IV. Volatility-scaled Historical Simulation	Wong Ka Nam2309349d
V. Garch(1,1) FHS	Fan Ho Yee 23089917d
Document related(Report, PowerPoint)	All

Resource list:

- https://finance.yahoo.com/quote/NVDA/
- https://finance.yahoo.com/quote/PLD/